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**FORM TP 2014234**

MAY/JUNE 2014

**CARIBBEAN EXAMINATIONS COUNCIL**  
**CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®**

**PHYSICS**

**UNIT 1 – Paper 02**

*2 hours 30 minutes*

**READ THE FOLLOWING INSTRUCTIONS CAREFULLY.**

1. This paper consists of SIX questions in TWO sections. Answer ALL questions.
2. For Section A, write your answers in the spaces provided in this booklet.
3. For Section B, write your answers in the spaces provided at the end of each question in this booklet.
4. All working **MUST** be shown.
5. You may use a silent, non-programmable calculator to answer questions, but you should note that the use of an inappropriate number of digits in answers will be penalized.

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**DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.**

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02138020/CAPE/2014

**NOTHING HAS BEEN OMITTED.**

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## LIST OF PHYSICAL CONSTANTS

|                                               |              |     |                                                       |
|-----------------------------------------------|--------------|-----|-------------------------------------------------------|
| Universal gravitational constant              | $G$          | $=$ | $6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$  |
| Acceleration due to gravity                   | $g$          | $=$ | $9.80 \text{ m s}^{-2}$                               |
| 1 Atmosphere                                  | $\text{Atm}$ | $=$ | $1.00 \times 10^5 \text{ N m}^{-2}$                   |
| Boltzmann's constant                          | $k$          | $=$ | $1.38 \times 10^{-23} \text{ J K}^{-1}$               |
| Density of water                              | $\rho_w$     | $=$ | $1.00 \times 10^3 \text{ kg m}^{-3}$                  |
| Specific heat capacity of water               | $C_w$        | $=$ | $4200 \text{ J kg}^{-1} \text{ K}^{-1}$               |
| Specific latent heat of vaporization of water | $l_v$        | $=$ | $2.26 \times 10^6 \text{ J kg}^{-1}$                  |
| Avogadro's number                             | $N_A$        | $=$ | $6.02 \times 10^{23} \text{ per mole}$                |
| Molar gas constant                            | $R$          | $=$ | $8.31 \text{ J K}^{-1} \text{ mol}^{-1}$              |
| Stefan-Boltzmann's constant                   | $\sigma$     | $=$ | $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ |
| Speed of light in free space                  | $c$          | $=$ | $3.00 \times 10^8 \text{ m s}^{-1}$                   |

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SECTION A

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

1. (a) Figure 1 shows a missile fired from the ground with a velocity of  $300 \text{ m s}^{-1}$  at an angle of inclination of  $40^\circ$ . The missile is in flight for 39 seconds.

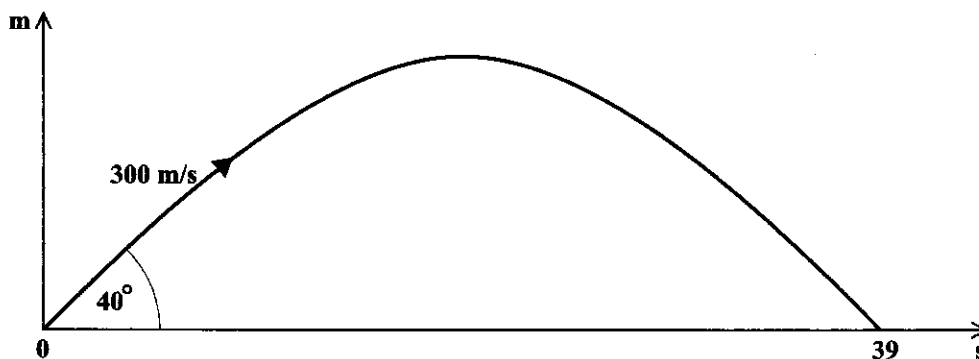


Figure 1

- (i) Calculate the initial horizontal component of the missile's velocity.

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[2 marks]

- (ii) By neglecting air resistance, determine the horizontal range of the missile.

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[2 marks]

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- (b) Figure 2, comprising six pictures labelled 1–6, represents a series of snapshots of a rubber ball in motion. The rubber ball was raised from the floor to a table of height 1.7 metres. It was then dropped from the table onto the floor and bounced up 1 metre.

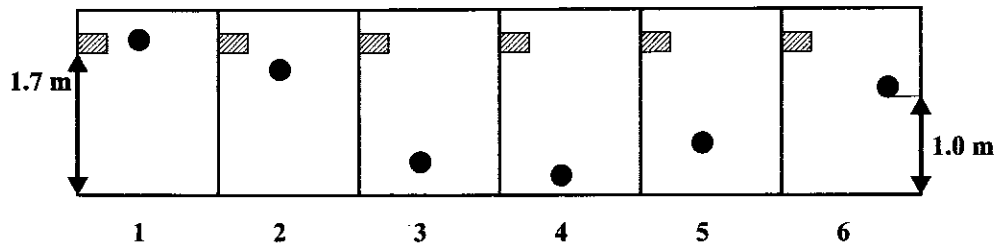


Figure 2

- (i) In which picture does the ball have maximum potential energy?
- \_\_\_\_\_ [1 mark]
- (ii) In which picture does the ball have maximum kinetic energy?
- \_\_\_\_\_ [1 mark]
- (iii) How much potential energy has the ball lost from Picture 1 to Picture 6, if the ball weighs 2 N?

[2 marks]

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- (c) Table 1 records the results of an experiment to determine the relationship between rebound height,  $H_{\text{rebound}}$ , and maximum height before rebound,  $H_{\text{before}}$ .

**TABLE 1**

| $H_{\text{before}}$ (m) | $H_{\text{rebound}}$ (m) |
|-------------------------|--------------------------|
| 1.70                    | 1.01                     |
| 1.00                    | 0.59                     |
| 0.59                    | 0.34                     |
| 0.36                    | 0.22                     |
| 0.21                    | 0.13                     |

On the grid provided on page 7, plot a graph of  $H_{\text{rebound}}$  versus  $H_{\text{before}}$  (starting at the origin). **[4 marks]**

- (d) (i) Use the graph in (c) above to deduce the equation relating  $H_{\text{rebound}}$  and  $H_{\text{before}}$ .

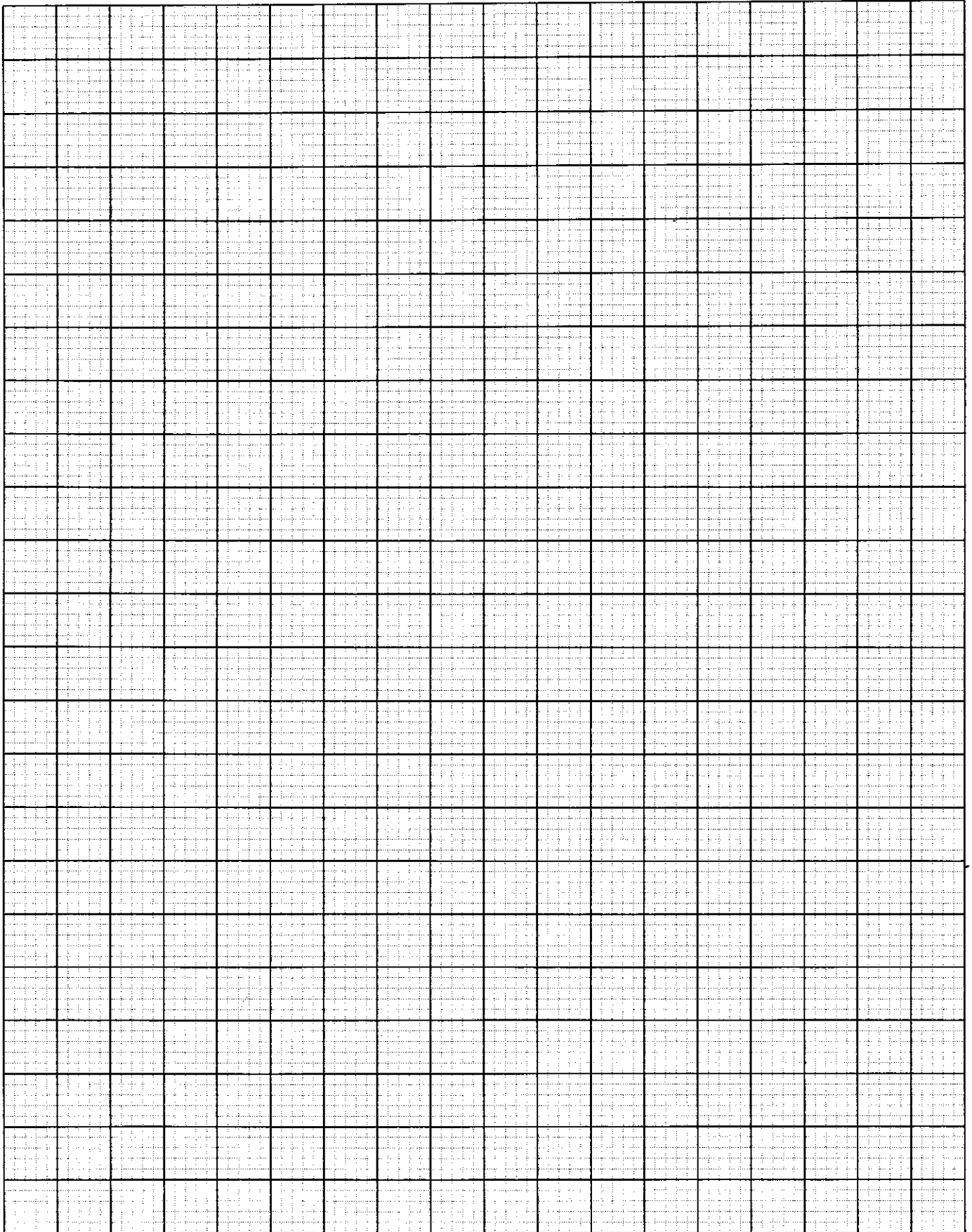
**[2 marks]**

- (ii) How high would the ball be expected to rebound if it were dropped from a height of 2.04 m?

**[1 mark]**

**Total 15 marks**

**GO ON TO THE NEXT PAGE**



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2. (a) (i) What is the approximate range of frequencies which can be heard by a young adult with normal hearing?

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**[1 mark]**

- (ii) A fifteen-year-old girl frequently listens to her iPod via a single earphone placed in her right ear. Describe an experiment which you could do in the physics laboratory to test whether the frequency response of her right ear is now different from that of her left ear.

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**[3 marks]**

- (b) The speed of ocean waves in deep water is given by

$$v = \sqrt{\frac{g\lambda}{2\pi}},$$

where  $g$  is the acceleration of free fall and  $\lambda$  is the wavelength of the waves. Show that  $T$ , the period of the waves, in terms of  $g$  and  $\lambda$  is given by

$$T = \sqrt{\frac{2\pi\lambda}{g}}.$$

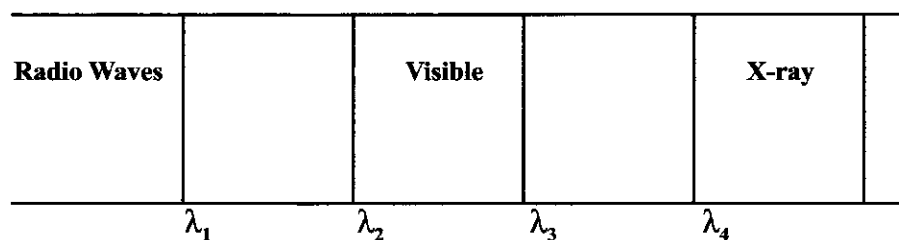
**[4 marks]**

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- (c) Calculate the value of  $T$  when the wavelength is 0.8 m.

[2 marks]

- (d) (i) Complete the diagram below to show the different regions of the electromagnetic spectrum.



Decreasing wavelength  $\longrightarrow$

[2 marks]

- (ii) Write down approximate values for the following:

$\lambda_1$  \_\_\_\_\_  
 $\lambda_2$  \_\_\_\_\_  
 $\lambda_3$  \_\_\_\_\_  
 $\lambda_4$  \_\_\_\_\_

[3 marks]

**Total 15 marks**

GO ON TO THE NEXT PAGE

3. In an experiment, a glass tube of cross sectional area  $6 \text{ mm}^2$  is used to study the relationship between volume and temperature of a fixed mass of gas at constant pressure. Table 2 shows some of the results of the experiment.

**TABLE 2**

| Temperature T<br>(°C) | Length of Gas<br>Column (mm) | Volume<br>V (mm <sup>3</sup> ) |
|-----------------------|------------------------------|--------------------------------|
| -35                   | 83                           |                                |
| -3                    | 89                           |                                |
| 27                    | 103                          |                                |
| 57                    | 110                          |                                |
| 87                    | 120                          |                                |

- (a) Complete Table 2 by inserting the missing values. **[2 marks]**
- (b) (i) Using the data from the table, plot on the grid provided on page 11, a graph of volume against temperature. Use a scale which shows  $-325^\circ\text{C}$  to  $+100^\circ\text{C}$  on the temperature axis and begin at  $0 \text{ mm}^3$  on the volume axis. **[4 marks]**
- (ii) From your graph, determine the temperature at which the volume is zero.

**[2 marks]**

- (c) If the gas is cooled enough, would its volume actually become zero? Suggest a reason for your answer.

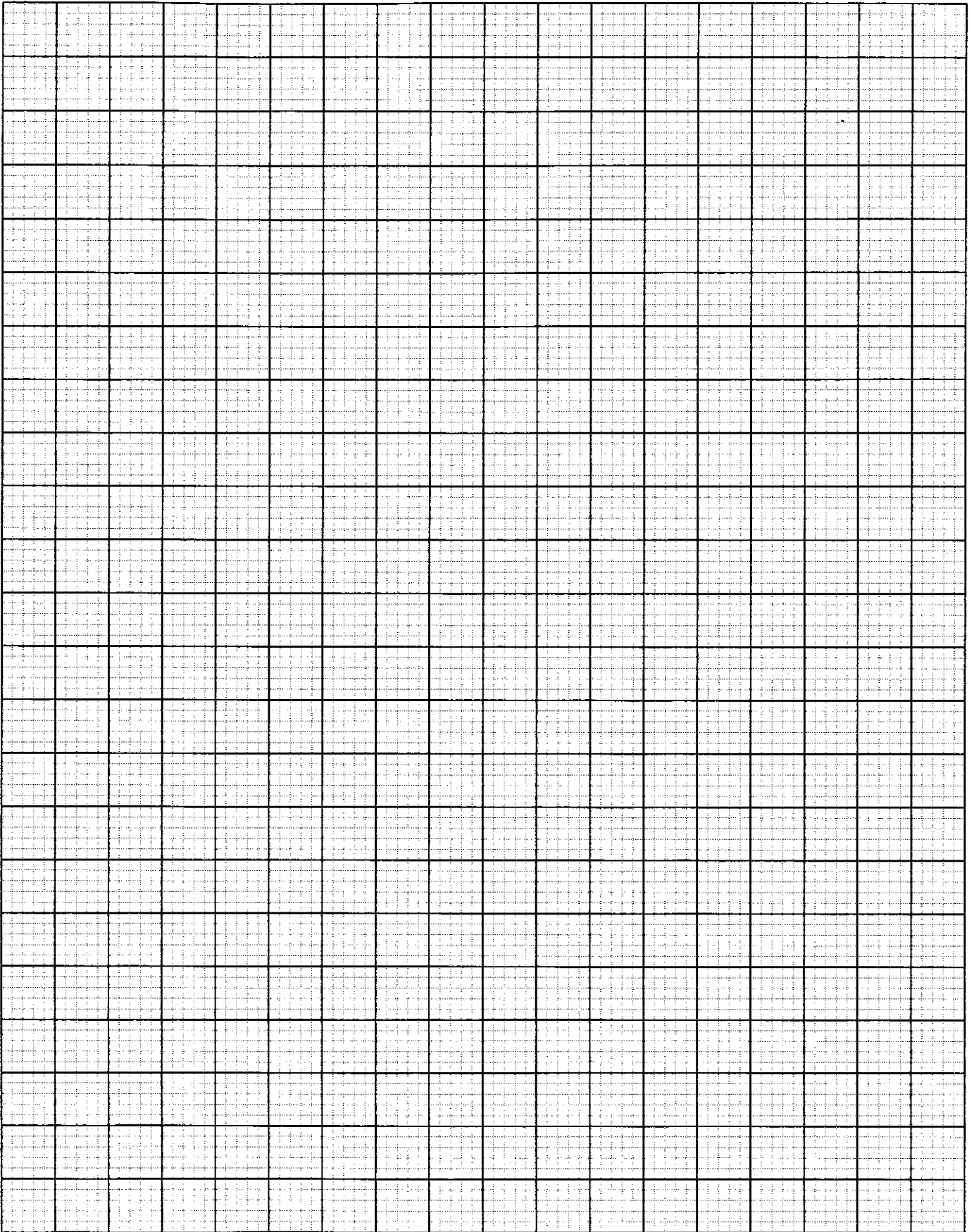
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**[2 marks]**

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- (d) Determine temperature, in kelvin, at which the length of the gas column would be 70 mm.

[3 marks]

- (e) (i) Write the equation of state for an ideal gas.

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[1 mark]

- (ii) Outline how this equation is applied in the experiment on page 10.

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[1 mark]

**Total 15 marks**

**NOTHING HAS BEEN OMITTED**

## SECTION B

Answer ALL questions.

Write your answers in the spaces provided at the end of each question.

4. (a) (i) Explain why a particle moving with constant speed along a circular path has an acceleration. **[2 marks]**
- (ii) The value of radial acceleration is given by the expression  $v^2/r$ , where  $v$  is the speed and  $r$  is the radius of the path. Using base units, show that this expression is dimensionally correct. **[2 marks]**
- (b) A racing car of mass 500 kg starts from rest and accelerates at  $6.0 \text{ m s}^{-2}$  along a straight horizontal road for a distance of 150 m. It then enters, at constant speed, a horizontal circular curve of radius 200 m.
- What is the magnitude **and** direction of the resultant horizontal force acting on the racing car while it is travelling around the curve? **[4 marks]**
- (c) A stone of mass 500 g is attached to a string of length 50 cm which will break if the tension exceeds 20 N. The stone is whirled in a vertical circle. The angular speed is slowly increased until the string breaks.
- (i) Draw a diagram to show the forces acting on the stone in a vertical circle when it is at its
- a) highest point **[1 mark]**
- b) lowest point. **[1 mark]**
- (ii) In what position along the vertical circular path is the string MOST likely to break? Suggest a reason for your answer. **[2 marks]**
- (iii) At what angular speed will the spring break? **[3 marks]**

**Total 15 marks**

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**Write the answer to Question 4 here.**

(a) (i)

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(ii)

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(b)

**Write the answer to Question 4 here.**

(c) (i)  $a - b$

(ii)

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(iii)

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5. (a) In order for waves to produce an observable interference pattern, they must be coherent. Explain what is meant by the term 'coherent' when used in this context. [1 mark]
- (b) A laser beam is incident on two narrow slits separated by a distance,  $a$ . The slits are at a distance,  $D$ , from a screen.
- (i) With the aid of a diagram, show that the distance,  $y$ , between the centre of the interference fringe pattern and the first bright fringe is given by
- $$y = \frac{\lambda D}{a}. \quad [5 \text{ marks}]$$
- (ii) With a pair of slits, spacing 0.20 mm, at a distance of 0.50 m from the screen, the bright fringes are 1.4 mm apart. What is the wavelength, in nm, of the light produced by the laser? [2 marks]
- (c) A parallel beam of sodium light is incident normally on a diffraction grating. The angle between the two first order spectra on either side of the normal is  $27.7^\circ$ . Assuming that the wavelength of the light is  $5.89 \times 10^{-7}$ , find
- (i) the number of rulings per mm on the grating [4 marks]
- (ii) the number of spectra obtained on either side of the normal. [3 marks]

**Total 15 marks**

**Write the answer to Question 5 here.**

(a)

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**Write the answer to Question 5 here.**

(b) (i)

(ii)

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(c) (i)

(ii)

6. (a) Figure 3 shows an example of a solar panel used in the Caribbean for heating domestic water. Discuss how the principles of conduction, convection, radiation and the greenhouse effect are applied in the design of this type of solar panels. [7 marks]

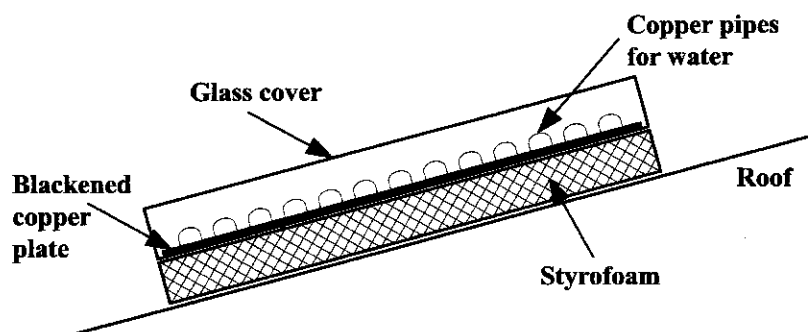


Figure 3

- (b) The wall of a cold store is constructed from two layers of brick with a foam-filled cavity between as shown in Figure 4. Each of the layers of the brick walls is 10 cm thick and the foam cavity is 5 cm wide. On a typical warm day, the temperature outside is 30 °C while the temperature inside the cold room is kept at -5 °C.

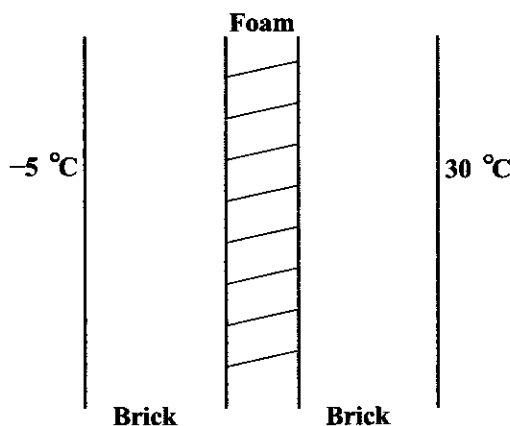


Figure 4. Cold store

- (i) Sketch a graph to show how the temperature would vary across the wall. [3 marks]
- (ii) The thermal conductivity of the brick is  $0.48 \text{ W m}^{-1} \text{ K}^{-1}$  and that of the foam is  $0.016 \text{ W m}^{-1} \text{ K}^{-1}$ . Calculate the thickness of brick which would have the same insulating effect as 5 cm of foam. [2 marks]
- (iii) Determine the rate at which heat energy would pass through each square metre of this composite wall. [3 marks]

Total 15 marks

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**Write the answer to Question 6 here.**

(a)

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**Write the answer to Question 6 here.**

**(b) (i)**

**(ii)**

(iii)

**END OF TEST**

**IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.**